

实验名称 实验1 生物分子数据库使用

一、实验内容及结果

1、以 *Streptomyces lividans* 1326 为关键词搜索 NCBI 数据库，获得基因组，分别以.gb 和 .fasta 格式保存；以 xlnA 为关键词搜索基因，下载来自 *Aspergillus clavatus* NRRL 1 的文件，将.fna 格式的文件保存；

2、以 1tux 为关键词搜索 PDB 数据库，下载 1tux 的 fasta 格式的文件以及.pdb 文件，显示蛋白质三维结构（可通过以下任何一个程序）；

- PyMol: <https://pymol.org/>
- Chimera: <https://www.cgl.ucsf.edu/chimera/>
- PMV: <http://www.scripps.edu/~sanner/python/>
- Coot: <https://www2.mrc-lmb.cam.ac.uk/personal/pemsley/coot/>
- CCP4mg: <http://www.ccp4.ac.uk/MG/>
- mmLib: <http://pymmlib.sourceforge.net/>
- VMD: <https://www.ks.uiuc.edu/Research/vmd/>
- MMTK: <http://dirac.cnrs-orleans.fr/MMTK/>

常用 pymol

3、以 Melanoma(黑色素瘤)为关键词，搜索 KEGG 数据库，获得人的 Melanoma 涉及的代谢途径，保存为 .png 文件；

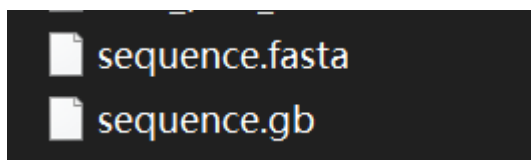
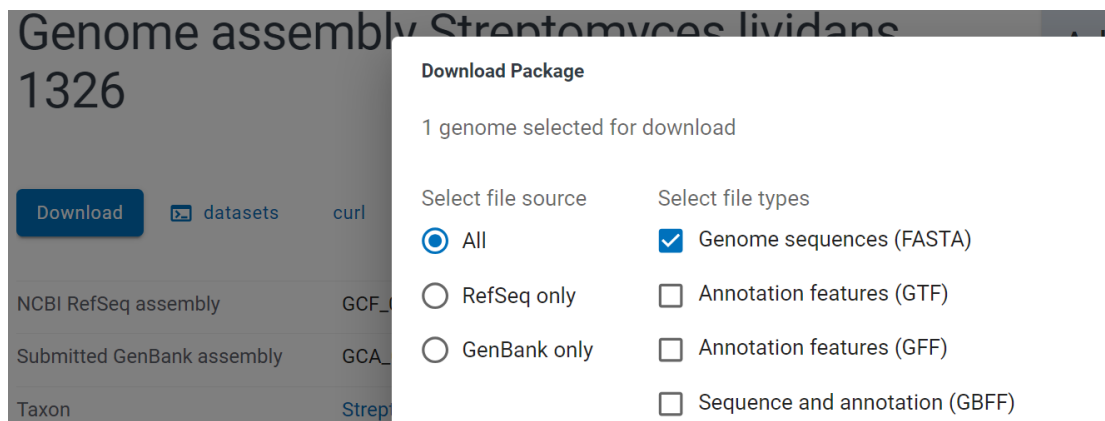
4、将第 3 步 Melanoma 代谢途径中的蛋白质输入 string 数据库，获得蛋白质-蛋白质相互作用（PPI）网络，并对网络进行分析。

源代码：(如果有)

运行结果 (截图):

Q1:

Streptomyces lividans 1326 基因组:



文件:

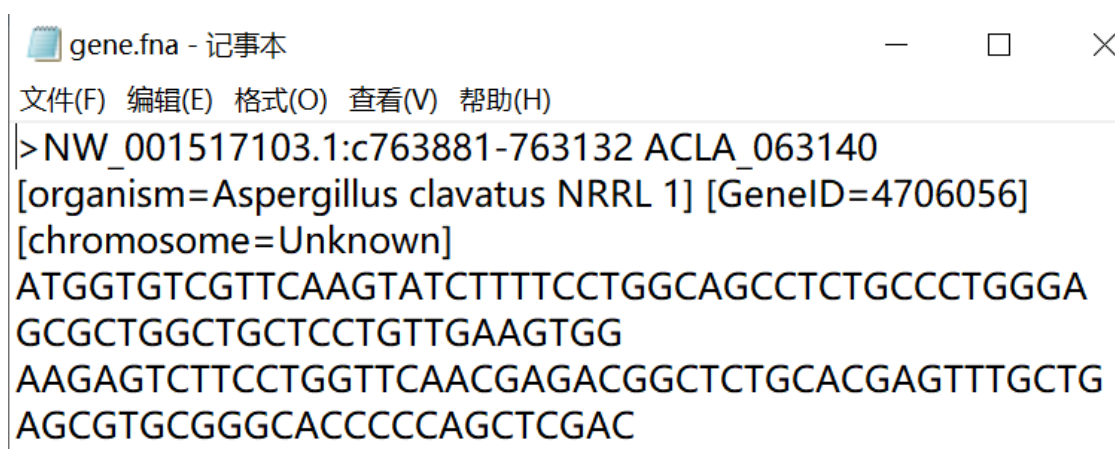
xlnA 基因:

ACLA_063140 endo-1,4-beta-xylanase (XlnA), putative [*Aspergillus clavatus* NRRL 1]

Gene ID: 4706056, updated on 28-Dec-2023

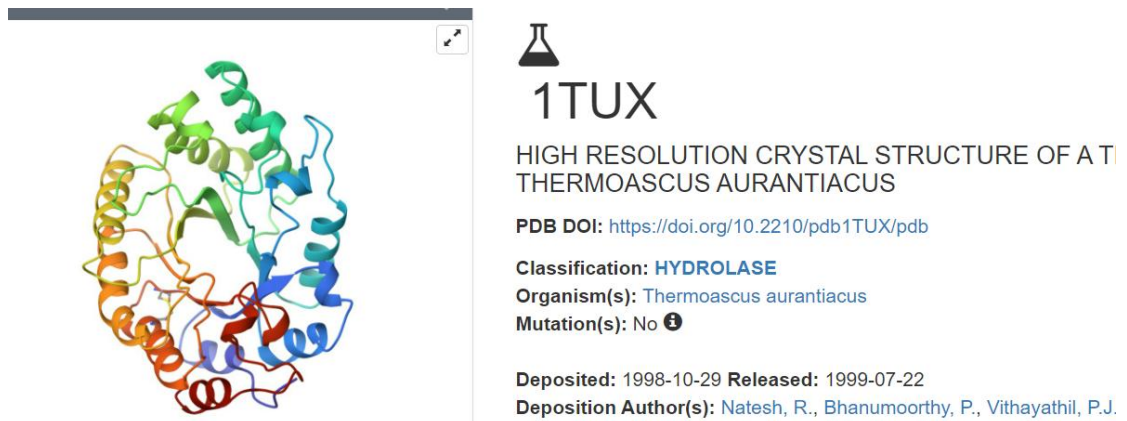
[Download Datasets](#)

文件:

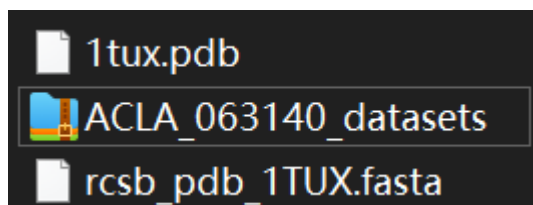


Q2:

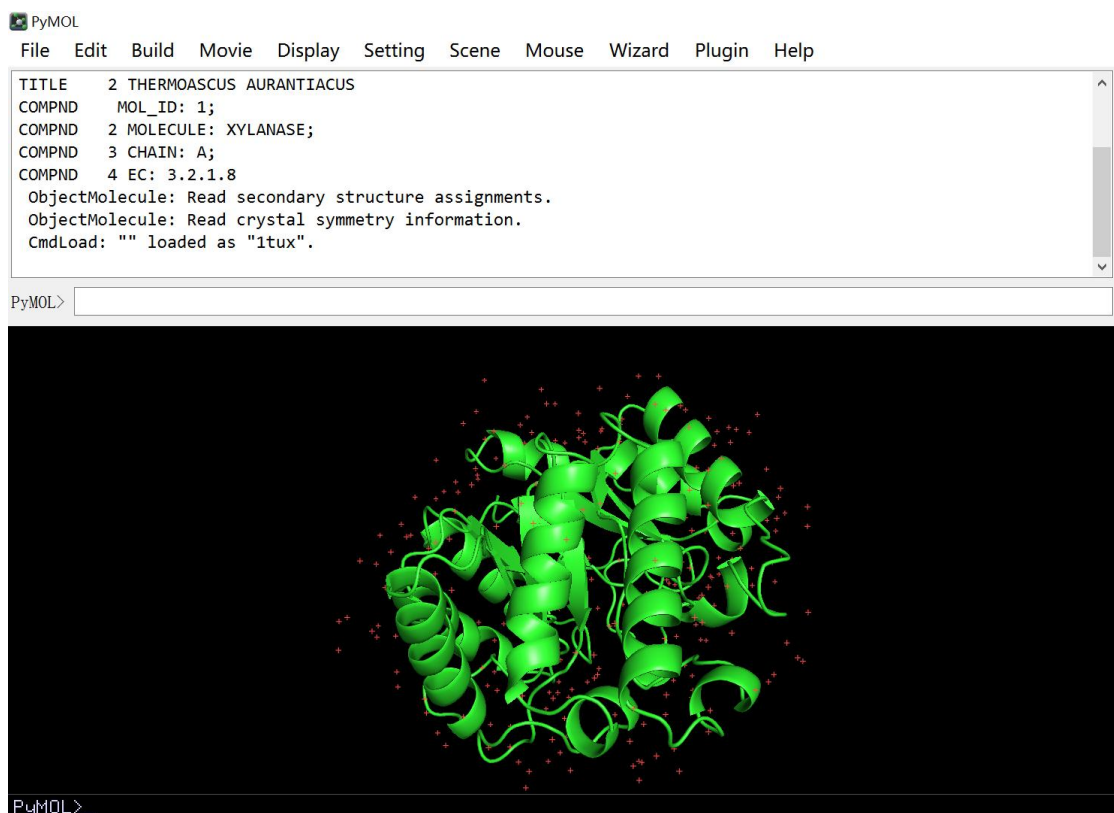
1tux:



文件:



Pymol 蛋白质三维结构:



Q3:

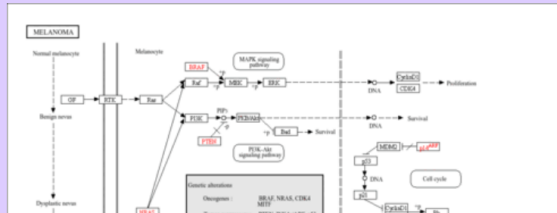
Entry	map05218	Pathway
Name	Melanoma	

Description

Melanoma is a form of skin cancer that has a poor prognosis and which is on the rise in Western populations. Melanoma arises from the malignant transformation of pigment-producing cells, melanocytes. The only known environmental risk factor is exposure to ultraviolet (UV) light and in people with fair skin the risk is greatly increased. Melanoma pathogenesis is also driven by genetic factors. Oncogenic NRAS mutations activate both effector pathways Raf-MEK-ERK and PI3K-Akt. The Raf-MEK-ERK pathway may also be activated via mutations in the BRAF gene. The PI3K-Akt pathway may be activated through loss or mutation of the inhibitory tumor suppressor gene PTEN. These mutations arise early during melanoma pathogenesis and are preserved throughout tumor progression. Melanoma development has been shown to be strongly associated with inactivation of the p16INK4a/cyclin dependent kinases 4 and 6/retinoblastoma protein (p16INK4a/CDK4, 6/pRb) and p14ARF/human double minute 2/p53 (p14ARF/HMD2/p53) tumor suppressor pathways. MTTF and TP53 are implicated in further melanoma progression.

Class Human Diseases; Cancer: specific types
BRITE hierarchy

map05218 Melanoma

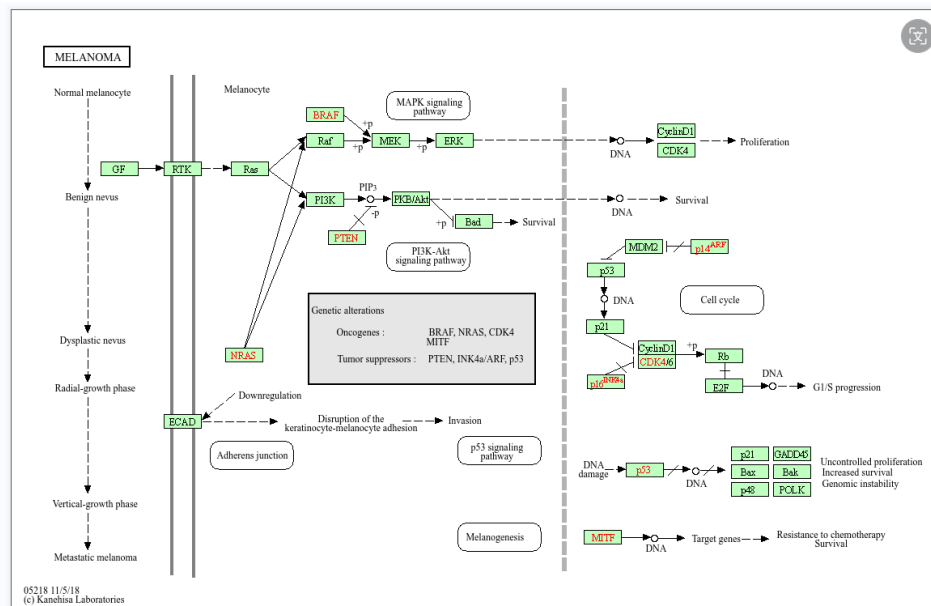


Pathway map

文件:

hsa05218_20240510_202448

map05218_20240510_195156

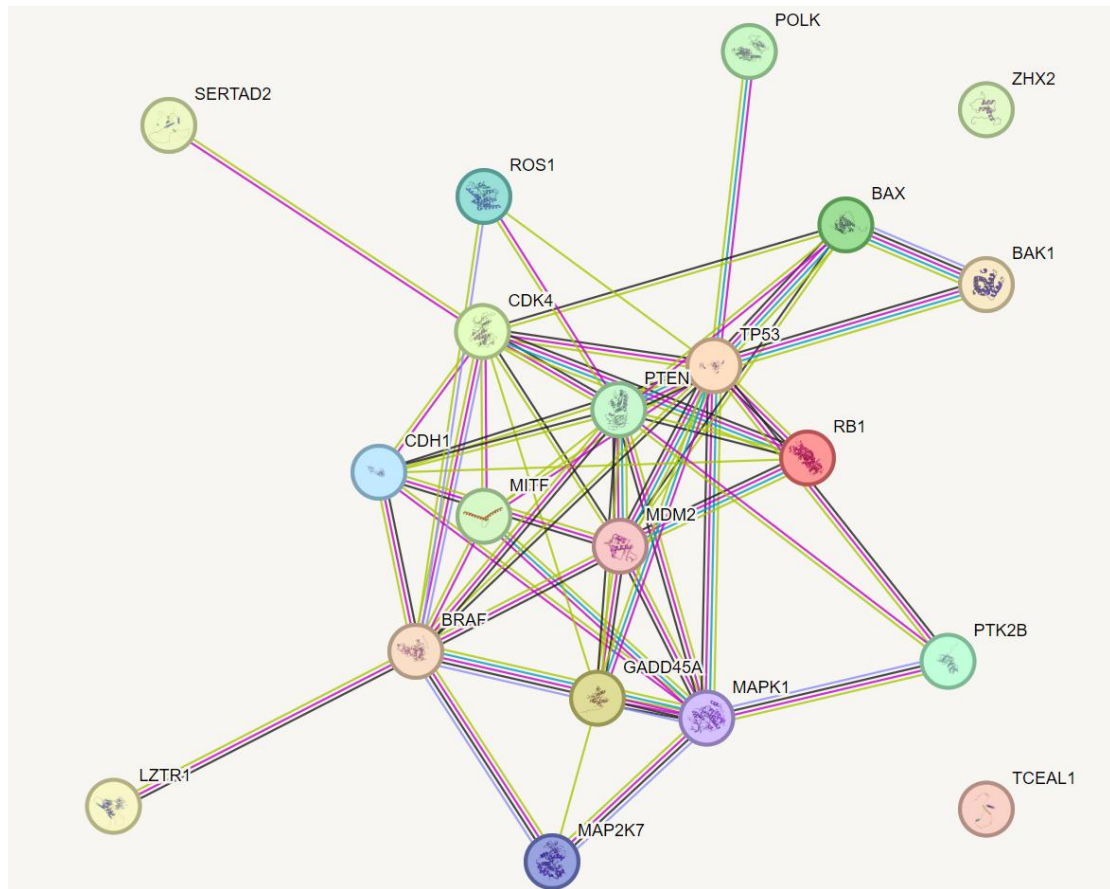


05218 11/5/18
(c) Kanehisa Laboratories

[全部图片](#)

Q4:

PPI 网络:



分析:

Network Stats

number of nodes: 20
number of edges: 53
average node degree: 5.3
avg. local clustering coefficient: 0.695

expected number of edges: 17
PPI enrichment p-value: $8.76e-13$
your network has significantly more interactions than expected (what does that mean?)

Functional enrichments in your network

[explain columns](#)

Biological Process (Gene Ontology)				
GO-term	description	count in network	strength	false discovery rate
GO:0002352	B cell negative selection	2 of 2	2.99	0.00094
GO:0072717	Cellular response to actinomycin D	2 of 4	2.69	0.0018
GO:0048597	Post-embryonic camera-type eye morphogenesis	2 of 4	2.69	0.0018
GO:1902262	Apoptotic process involved in blood vessel morphogenesis	2 of 5	2.6	0.0023
GO:0034349	Glial cell apoptotic process	2 of 5	2.6	0.0023

(more ...)

Molecular Function (Gene Ontology)				
GO-term	description	count in network	strength	false discovery rate
GO:0051400	BH domain binding	2 of 10	2.29	0.0207
GO:0004708	MAP kinase kinase activity	3 of 17	2.24	0.0050
GO:0008353	RNA polymerase II CTD heptapeptide repeat kinase activity	2 of 12	2.22	0.0256
GO:0004707	MAP kinase activity	2 of 15	2.12	0.0342
GO:0097718	Disordered domain specific binding	3 of 34	1.94	0.0056

(more ...)